

Question 1

a) Mean, $\bar{x} = \frac{\sum x}{N}$

$$= \frac{430 + 445 + 455 + 435 + 440 + 450 + 455 + 460 + 435 + 445}{10}$$

$$= 445 \text{ MPa}$$

b) Median = $\left(\frac{N+1}{2}\right)^{\text{th}}$ observation

$$= \left(\frac{10+1}{2}\right)^{\text{th}} \text{ observation}$$

$$= 5.5^{\text{th}} \text{ observation}$$

The median is the average of the 5th and 6th observations in the sorted data.
Sorted data: 430, 435, 435, 440, 445, 445, 450, 455, 455, 460

$$5^{\text{th}} \text{ value} = 445 \text{ MPa}$$

$$6^{\text{th}} \text{ value} = 445 \text{ MPa}$$

$$\text{So, Median} = \frac{445 + 445}{2}$$

$$= 445 \text{ MPa}$$

c) There are three values that occur most frequently: 435, 445, and 455.

$$\text{Mode} = 435, 445, \text{ and } 455$$

This means the data is multi modal distribution, as it has three modes.

The mean, median, and mode they are all near 445. This shows the strength numbers are consistent. The mean is the best number to use because there are no extreme values pulling the average away from the middle.

Question 2

a) Range = Largest Value - Smallest Value

$$= 460 - 430$$

$$= 30 \text{ MPa}$$

b) Variance, $s^2 = \frac{\sum (xi - \bar{x})^2}{n-1}$

Where $\bar{x} = 445$ MPa

$$\begin{aligned}\therefore \sum (xi - \bar{x})^2 &= (430 - 445)^2 + (435 - 445)^2 + (435 - 445)^2 + (440 - 445)^2 + (445 - 445)^2 + (445 - 445)^2 \\ &\quad + (450 - 445)^2 + (455 - 445)^2 + (455 - 445)^2 + (460 - 445)^2 \\ &= 900 \\ \therefore \text{Variance, } s^2 &= \frac{900}{10-1} \\ &= 100 \text{ MPa}^2\end{aligned}$$

c) Standard Deviation, $s = \sqrt{\text{Variance}}$

$$\begin{aligned}&= \sqrt{100} \\ &= 10 \text{ MPa}\end{aligned}$$

The standard deviation is small relative to the mean, indicating consistent material quality across the measured rods.

Question 3

A)

Sorted data: 430, 435, 435, 440, 445, 445, 450, 455, 455, 460

Here,

First Quartile, $Q_1 = (i(\frac{N+1}{4}))^{\text{th}} \text{ observation} = (\frac{10+1}{4})^{\text{th}} \text{ observation}$

$$= 2.75^{\text{th}} \text{ observation}$$

$$= 2^{\text{nd}} \text{ observation} + (3^{\text{rd}} \text{ observation} - 2^{\text{nd}} \text{ observation}) \times 0.75$$

$$= 435 + (435 - 435) \times 0.75$$

$$= 435$$

Median (Q_2) = $(i(\frac{N+1}{4}))^{\text{th}} \text{ observation} = (2 \times \frac{10+1}{4})^{\text{th}} \text{ observation}$

$$= 5.5^{\text{th}} \text{ observation}$$

$$\begin{aligned}
&= 5^{\text{th}} \text{ observation} + (6^{\text{th}} \text{ observation} - 5^{\text{th}} \text{ observation}) \times 0.5 \\
&= 445 + (445 - 445) \times 0.5 \\
&= 445
\end{aligned}$$

$$\begin{aligned}
\text{Third Quartile, } Q_3 &= (i(\frac{N+1}{4}))^{\text{th}} \text{ observation} = (3 \times \frac{10+1}{4})^{\text{th}} \text{ observation} \\
&= 8.25^{\text{th}} \text{ observation} \\
&= 8^{\text{th}} \text{ observation} + (9^{\text{th}} \text{ observation} - 8^{\text{th}} \text{ observation}) \times 0.25 \\
&= 455 + (455 - 455) \times 0.25 \\
&= 455
\end{aligned}$$

Minimum = 430

Maximum = 460

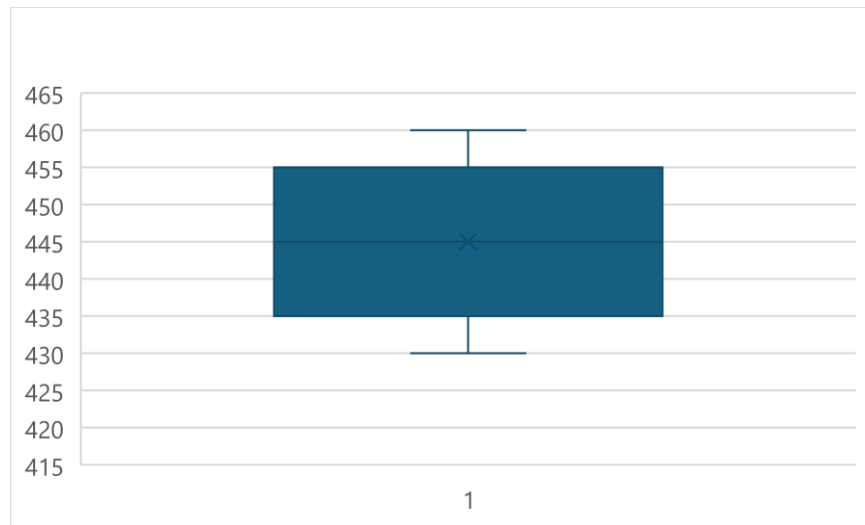


Figure 1: Box Plot

There are no outliers in this dataset. This means all the measured steel rods exhibit tensile strength within the expected range

B)

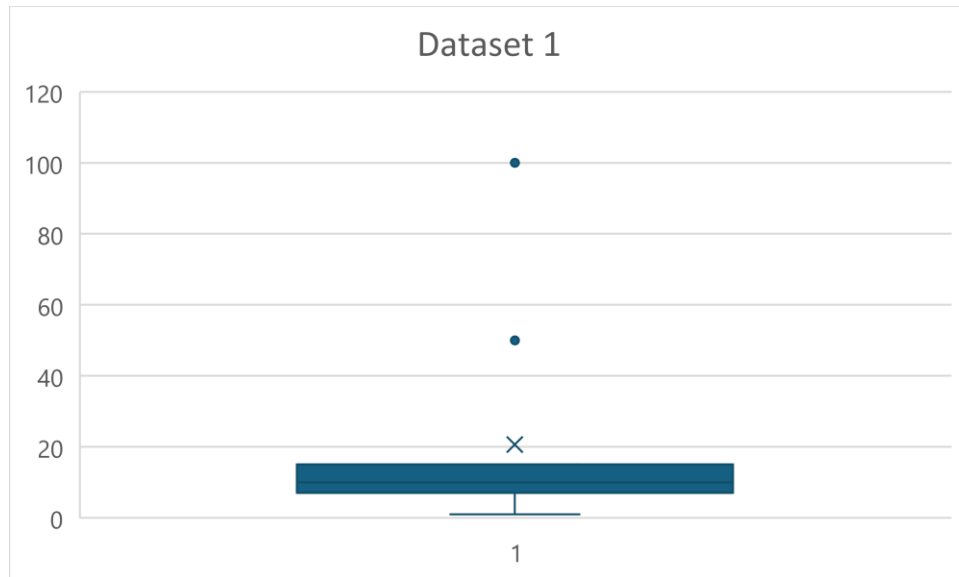


Figure 2: Box Plot

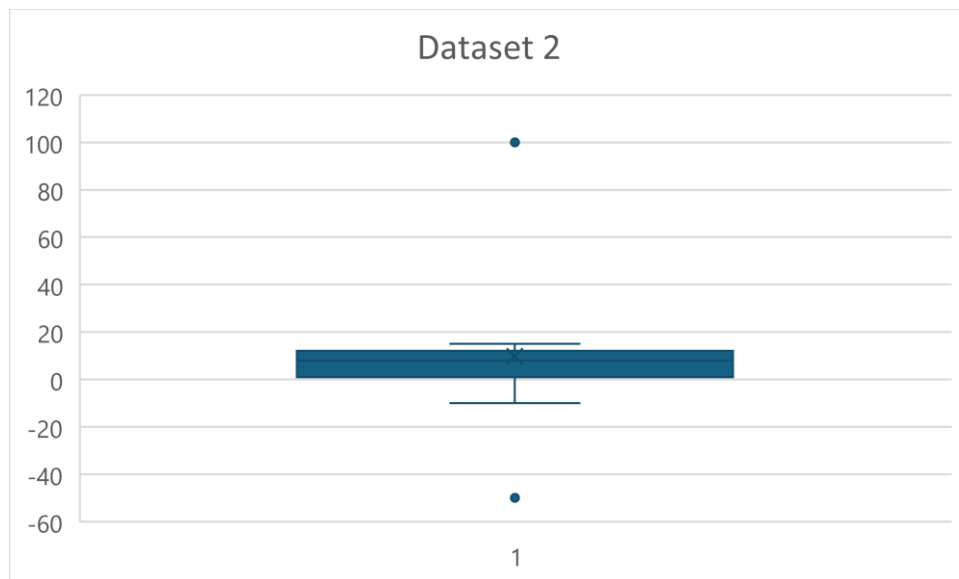


Figure 3: Box Plot

Dataset 1 and Dataset 2 exhibit several key differences in their distribution as shown by the box plots. Dataset 1's distribution is strongly skewed to the right (positively skewed) because the median (the line in the box, around 9) is close to the first quartile (Q_1 , the left edge of the box, around 7), and the upper whisker is long, with two high outliers at 50 and 100. The box itself is narrow, indicating that the middle 50% of the data points are tightly clustered between 7 and 15. In contrast, Dataset 2 has a wider box, showing that its middle 50% of values are more spread out (from Q_1 at 1 to Q_3 at 12). While its median (around 8) is similar to Dataset 1, Dataset 2 is notable for the extreme negative outlier at -50, which drastically extends the overall range to the

left. Both datasets share the same maximum value of 100, though it is an outlier in both cases. In summary, Dataset 1's spread is dominated by high extreme values, while Dataset 2's spread is significantly influenced by a single, very low extreme value.

Question 4:

Class Interval (Ω)	Class Mid (x_i)	Frequency (f_i)	$f_i x_i$	$\frac{f_i}{x_i}$	Cumulative Frequency (CF)
95-99	97	5	485	0.051545	5
100-104	102	9	918	0.088235	14
105-109	107	11	1177	0.102804	25
110-114	112	13	1456	0.116071	38
115-119	117	7	819	0.059803	45
120-124	122	3	366	0.024590	48
125-129	127	2	254	0.015748	50
Total		n=50	$\sum f_i x_i = 5475$	$\sum \frac{f_i}{x_i} \approx 0.458796$	

a) Arithmetic mean, $\bar{x} = \frac{\sum f_i x_i}{n} = \frac{5475}{50} = 109.5 \Omega$

Geometric mean, $G = \text{antilog}(\frac{1}{n} \sum f_i \log x_i)$

$$= \text{antilog}(\frac{1}{50} (5 \times \log(97) + 9 \times \log(102) + 11 \times \log(107) + 13 \times \log(112) + 7 \times \log(117) + 3 \times \log(122) + 2 \times \log(127)))$$

$$= 109.23 \Omega$$

Harmonic mean, $H.M = \frac{n}{\sum \frac{f_i}{x_i}} = \frac{50}{0.458796} = 108.98 \Omega$

b)

$$\text{Median} = L + \frac{\frac{N}{2} - p.c.f.}{f} \times i$$

Median class: $CF \geq (\frac{N}{2})^{\text{th}}$ observation

$$= (\frac{50}{2})^{\text{th}} \text{ observation} = 25^{\text{th}} \text{ observation}$$

∴ Median class is 105-109.

$$\begin{aligned}\therefore \text{Median} &= 105 + \frac{\frac{50}{2} - 14}{11} \times 4 \\ &= 109 \Omega\end{aligned}$$

c)

$$\text{Mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times i$$

Modal class is 110 – 114

$$\begin{aligned}\therefore \text{Mode} &= 110 + \frac{13-11}{(13-11)+(13-7)} \times 4 \\ &= 111\end{aligned}$$

d)

Class Interval (Ω)	Frequency, f_i	Class Mid, x_i	$f_i x_i$
95 - 99	5	97	485
100 – 104	9	102	918
105 – 109	11	107	1177
110 – 114	13	112	1456
115 – 119	7	117	819
120 – 124	3	122	366
125 – 129	2	127	254
Total	N = 50		$\sum f_i x_i = 5475$

$$\begin{aligned}\text{Mean, } \bar{x} &= \frac{\sum fixi}{\sum fi} \\ &= \frac{5475}{50} \\ &= 109.5 \, \Omega\end{aligned}$$

$$\begin{aligned}\text{Now, } S &= \sqrt{\frac{\sum fi(xi - \bar{x})^2}{n-1}} \\ &= \sqrt{\frac{5(97-109.5)^2 + 9(102-109.5)^2 + 11(107-109.5)^2 + 13(112-109.5)^2 + 7(117-109.5)^2 + 3(122-109.5)^2 + 2(127-109.5)^2}{50-1}} \\ &= 7.7 \, \Omega\end{aligned}$$

$$\begin{aligned}\text{So, Coefficient of variation, C.V} &= \frac{7.7}{109.5} \times 100 \\ &= 7.03\%\end{aligned}$$

The coefficient of variation tells us how uniform the resistors are. A low percentage means the resistance values don't spread out much and are consistent around the mean.